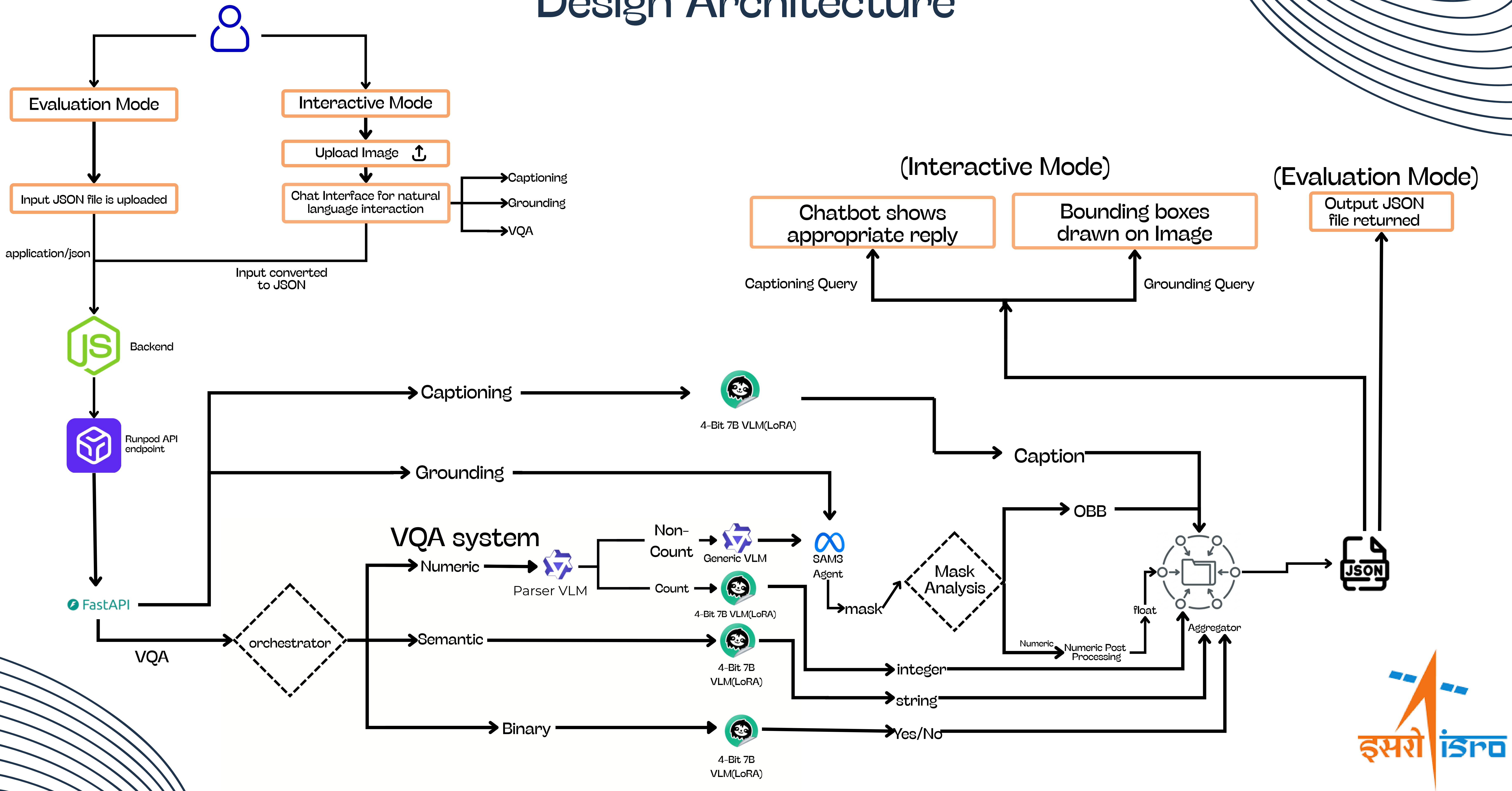
GeoNLI

Natural Language Interpretation of Satellite Imagery

Inter IIT Tech Meet 14.0

By Team_24

Design Architecture



Software Stack & Models

FastAPI
VLLM
Hugging Face
Unsloth

Qwen
SAM-3
PyTorch

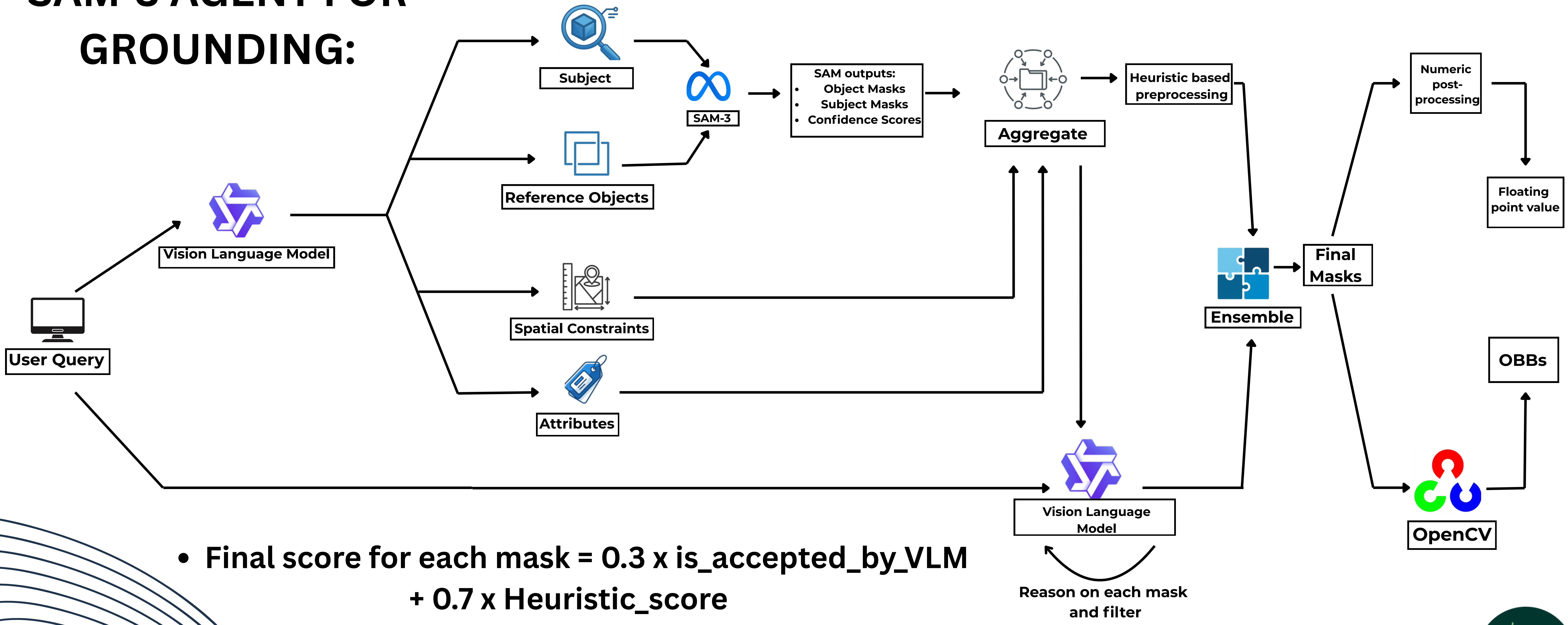
Vercel
Render
Docker
Runpod

Express JS
NodeJS
ReactJS

Runpod GPUs (2x A40 @0.80$/hr and 100 GB persistent volume @0.014$/hr)

Our Innovation (1/2)

SAM-3 AGENT FOR GROUNDING:

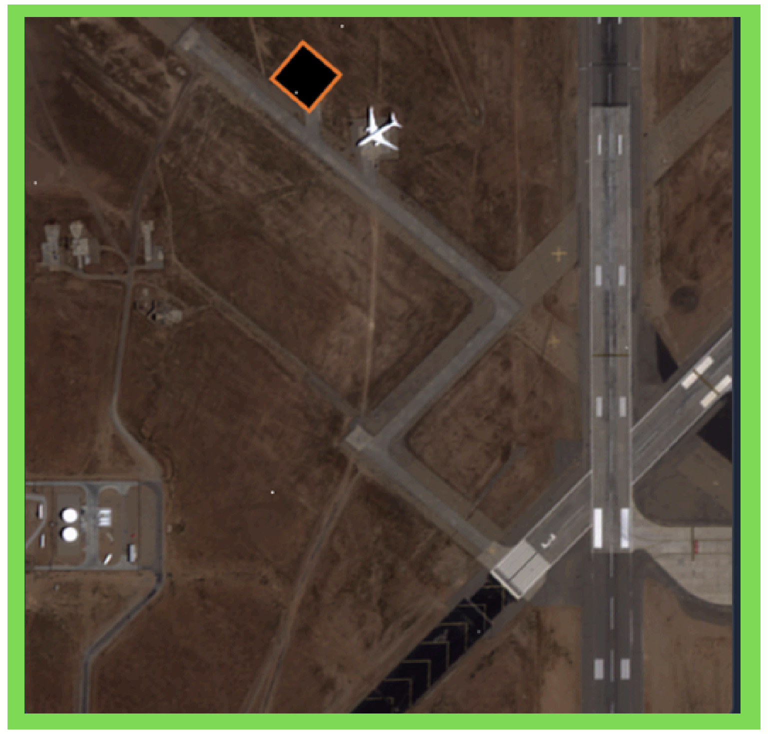


- Final score for each mask = 0.3 x is_accepted_by_VLM + 0.7 x Heuristic_score

Our Innovation (2/2)

PROMPT: "Locate and return oriented bounding boxes for the left aircraft"

- Black-Box Limitation: Relies purely on embedding similarity, often failing to interpret relative spatial terms like "left" or "second."
- Over-Detection: Detects the general class ("aircraft") but ignores the specific constraint, returning masks for all visible planes.
- Result: Cluttered output with low precision and high false positives.



- Structured Parsing: An MLLM decomposes the prompt into strict constraints (e.g., target: "aircraft", region: "left").
- Neuro-Symbolic Logic: Instead of guessing, we generate a full population of candidates and use Python scripts to mathematically filter them.
- Deterministic Accuracy: Coordinate-based sortinuarantees the model selects the exact "leftmost" object, eliminating spatial hallucinations.

Test Results

Qwen2.5-VL-7B-Instruct-bnb-4bit (Captioning)

0.68

BERT-BLEU4

Qwen2.5-VL-7B Instruct-bnb-4bit (Binary VQA)

0.91

Binary Exact Match

Qwen2.5-VL-7B Instruct-bnb-4bit (Semantic VQA)

0.85

BERT-BLEU4

Qwen2.5-VL-7B-Instruct-bnb-4bit (Numeric VQA)

0.73 / 1.0

Normalized Absolute Difference

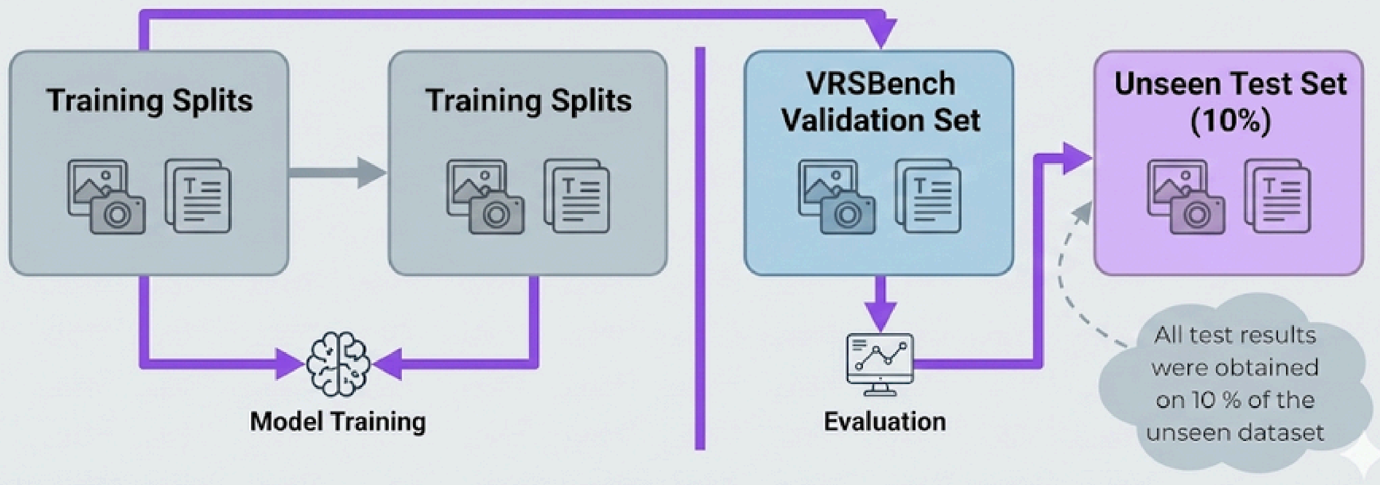
SAM3 (Grounding)

Mean IOU	0.43
Accuracy @ 0.5	0.57
Accuracy @ 0.7	0.40

Test Philosophy

Strict Data Isolation

We enforced zero data leakage by training exclusively on training splits and evaluating strictly on the VRSBench validation set to ensure true generalization.



Metric-Centric Optimization

Our fine-tuning and prompting strategies were explicitly tailored to maximize competition metrics (BERT-BLEU4, Exact Match & Penalty-IoU).

ABLATION-DRIVEN ARCHITECTURE

We systematically validated every architectural decision (e.g., adding SAM3 for grounding) by comparing Zero-shot vs. SFT performance to prove module necessity.

Limitations

Due to resource constraints, our final inference was done on 2 x A40.

Limited Remote Sensing domain knowledge in SAM3 leading to erroneous masks in low contrast or ambiguous structures.

The website cannot handle more than 3 concurrent sessions by multiple users.

Better handling of Multimodalities like SAR imagery, infrared imagery, etc. could have been possible.

Explore techniques other than explicit context aggregation to mitigate lack of spatial understanding in VLMs

Absence of grounded-reasoning training (DPO, GRPO, Attention Reweighting)

Thank You